FINITE MORPHISMS OVER QCQS BASES ARE LIMITS OF FINITE FINITE-PRESENTATION MORPHISMS

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Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Finite type is limit fp. Limits, Lemma lemma-finite-type-is-limit-finite-presentation (same chapter)
- Eventually finite. Limits, Lemma lemma-eventually-finite (same chapter)

Source reference. Original lemma 09ZQ: sources/stacks-chow/tags.txt, source lines 12608–12608. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09ZN: sources/stacks-chow/tags.txt, source lines 12606–12606. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Assume*

(1) *f is finite, and*
(2) *S quasi-compact and quasi-separated.*

Then X is a directed limit $X = \lim X_i$ where the transition maps are closed immersions and the objects X_i are finite and of finite presentation over S.

Proof. We may write $X = \lim X_i$ as in Lemma 09ZQ. Applying Lemma 09ZN we see that $X_i \to S$ is finite for large enough i. □

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.